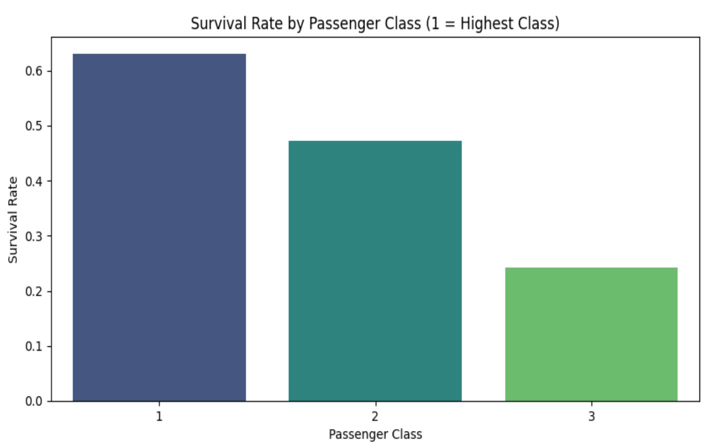
<u>**Task 2 Summary**</u>

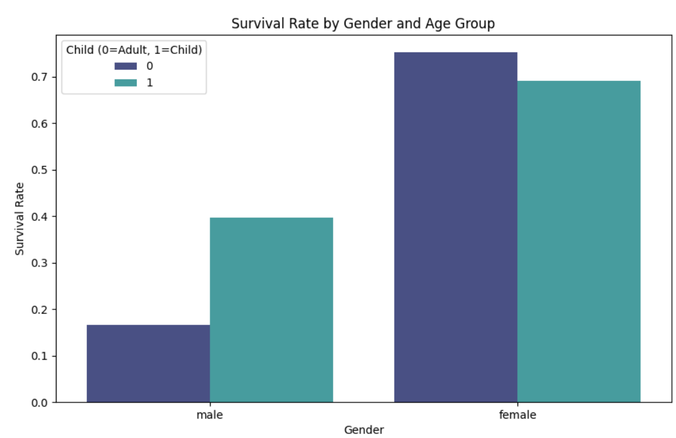
```
PS D:\Python Codes> & C:/Users/Fatmi/AppData/Local/Programs/Python/Python313/python.exe "d:/Python Codes/CodeAlphaInternship
========================================
Dataset Shape: (891, 15)

First 5 Rows:
   survived  pclass     sex   age  sibsp  parch     fare embarked  class    who  adult_male deck  embark_town alive  alone
0         0       3    male  22.0      1      0   7.2500        S  Third    man        True  NaN  Southampton    no  False
1         1       1  female  38.0      1      0  71.2833        C  First  woman       False    C    Cherbourg   yes  False
2         1       3  female  26.0      0      0   7.9250        S  Third  woman       False  NaN  Southampton   yes   True
3         1       1  female  35.0      1      0  53.1000        S  First  woman       False    C  Southampton   yes  False
4         0       3    male  35.0      0      0   8.0500        S  Third    man        True  NaN  Southampton    no   True

Data Types:
 survived          int64
pclass            int64
sex              object
age             float64
sibsp             int64
parch             int64
fare            float64
embarked         object
```

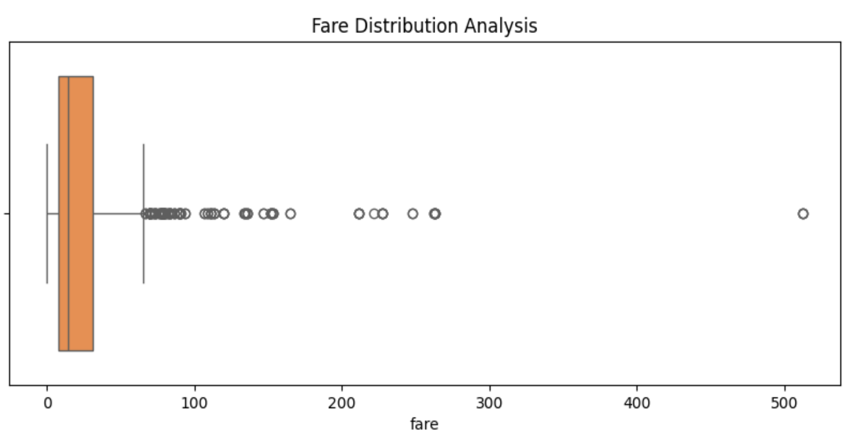
```
Missing Values:
 survived         0
pclass           0
sex              0
age            177
sibsp            0
parch            0
fare             0
embarked         2
class            0
who              0
adult_male       0
deck           688
embark_town      2
alive            0
alone            0
dtype: int64


==========================
After Cleaning:
```

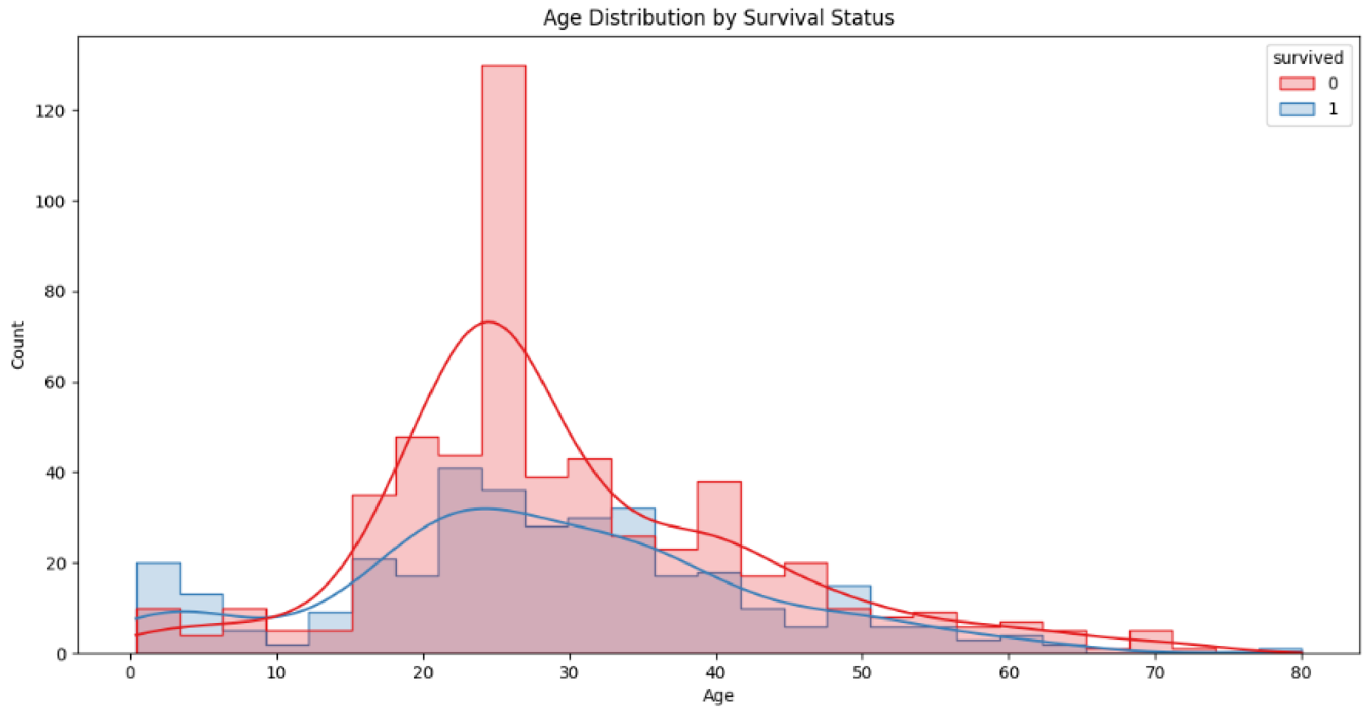
```
==============================
After Cleaning:
Missing Values:
 survived        0
pclass          0
sex             0
age             0
sibsp           0
parch           0
fare            0
embarked        0
class           0
who             0
adult_male      0
embark_town     2
dtype: int64
d:\Python Codes\CodeAlphaInter
```

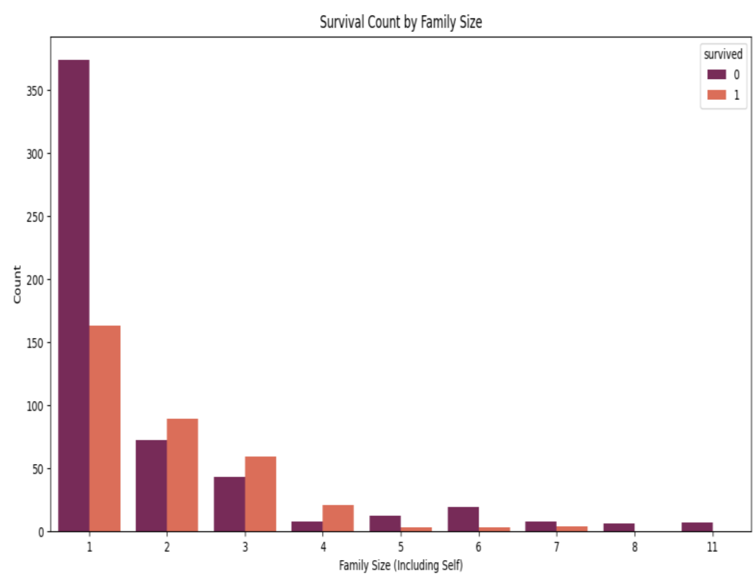
```
Top 5 Highest Fares:
         fare  pclass     sex   age embarked
258  512.3292       1  female  35.0        C
679  512.3292       1    male  36.0        C
737  512.3292       1    male  35.0        C
27   263.0000       1    male  19.0        S
88   263.0000       1  female  23.0        S

Hypothesis 1 (Class vs Survival): p-value = 0.0000
Hypothesis 2 (Age Impact): p-value = 0.0003

========================================
Key Insights (Calculated from Data):
- 1st Class passengers: 63.0% survival vs 3rd Class: 24.2%
- Female survival rate: 74.2% vs Male: 18.9%
- Children (<18): 54.0% survival vs Adults: 36.1% (p < 0.0001)
- Correlation between class and survival: -0.34 (Negative = Lower class numbers (1st) correlate with survival)
PS D:\Python Codes>
```

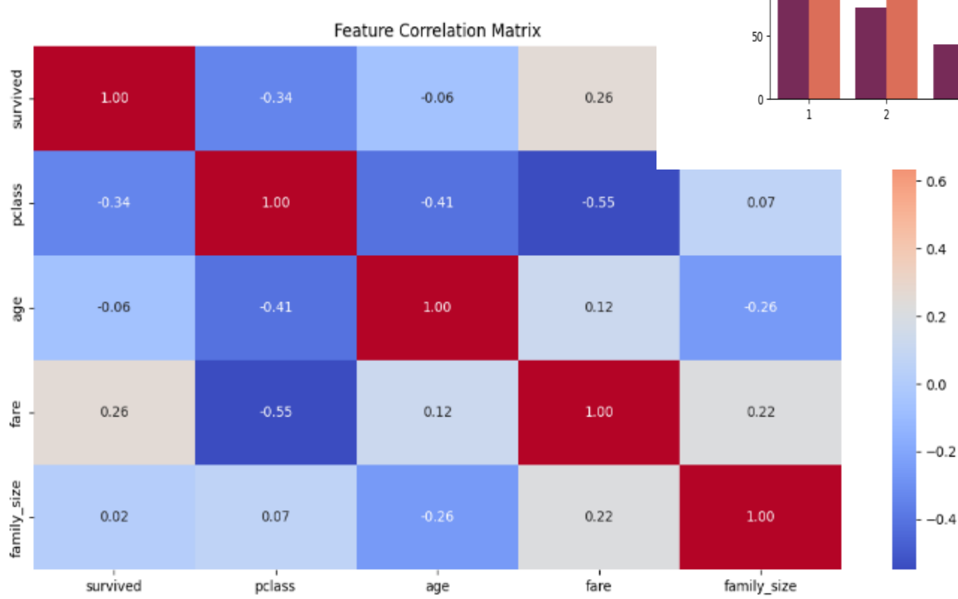







Age Distribution by Survival Status
survived
0
1
Count
Age
0
20
40
60
80
100
120
10
30
50
70


Survival Count by Family Size
survived
0
1
Count
Family Size (Including Self)
0
50
100
150
200
250
300
350
1
2
3
4
5
6
7
8
11


Feature Correlation Matrix
survived
pclass
age
fare
family_size
1.00
-0.34
-0.06
0.26
-0.41
-0.55
0.07
0.12
-0.26
0.22
0.02
0.6
0.4
0.2
0.0
−0.2
−0.4